DW1000 UWB Ranging Bias Characterisation

Three-Dataset Empirical Analysis

UWB RTLS Project — 2026-06-19

UWB RTLS Project

Abstract

We present an empirical characterisation of distance-dependent ranging bias in a Decawave DW1000–based ultra-wideband (UWB) real-time location system. Three datasets were collected on 2026-06-19 from two anchors across 13 distances (0.05–6 m), with 500 ranging samples per distance per session. The primary finding is that calibration error explains the dominant systematic offset: an anchor calibrated with a wrong reference distance shows a +228 mm higher mean bias than the same anchor after correct recalibration. After accurate antenna-delay calibration, the *residual* bias ranges from -96 mm to +130 mm across distances, with an irregular, non-monotonic profile that we attribute to room-specific multipath interference rather than hardware LDE physics. Session repeatability is excellent (RMS inter-session $\Delta \leq 8\,mm$); battery replacement has no detectable effect ($\Delta = -0.7\,mm$ at 4 m). Leave-one-out cross-validation of five correction models shows that a simple constant offset (RMSE $= 43.7\,mm$) outperforms all polynomial and spline alternatives. Cross-anchor generalisation of a polynomial correction fails badly (RMSE $= 95\,mm$ on an unseen anchor). These results do *not* support implementing a distance-dependent correction. Instead, the data suggests that calibration accuracy and multipath mitigation are the dominant levers.

1 Introduction

Ultra-wideband (UWB) time-of-flight ranging using the Decawave DW1000 is known to exhibit systematic distance-dependent bias beyond antenna-delay calibration error. Decawave application note APS011 attributes the dominant residual to a first-path detection (LDE) sensitivity that varies with received signal amplitude: at close range, high received power causes the leading-edge discriminator to trigger slightly early, reading shorter than the true distance; at long range, low SNR causes the LDE to miss the first-path peak and latch a later multipath arrival, reading longer.

The practical question for this system is: does this effect exist at measurable amplitude in our operating environment, and if so, does it warrant a correction model in the host pipeline?

This report answers that question from three datasets collected in a single measurement session. It deliberately avoids inheriting assumptions from earlier calibration analyses, treats each dataset on its own terms, and challenges each major conclusion against counter-evidence.

2 Experimental Setup

2.1 Hardware

All boards are Makerfabs "ESP32 UWB Pro with Display" units (Decawave DW1000, 64 MHz PRF, 110 kbps data rate, 2048-symbol preamble, reply delay $\tau_r = 5000\,\mu s$). Antenna delays were calibrated via a binary-search procedure using a 3.0 m reference distance.

2.2 Measurement protocol

1. Tag (0xF0) and a single anchor placed at a measured distance.
2. 500 consecutive DS-TWR range measurements collected per session.
3. Two sessions per distance, with re-positioning between sessions.
4. Data logged via UDP to `collect_bias_data.py` at $\approx 10\,Hz$.
5. 13 distances: 5, 50, 100, . . . , 600 cm.

2.3 Datasets

Three CSV files were collected; see Table 1.

Table 1: Dataset descriptions.

Key	Description
A2_run1	Anchor 2, calibrated previous evening. Collected next day (500 samples/distance, 2 sessions).
A1_fresh	Anchor 1, freshly calibrated at 3.0 m then swept immediately. Only 6 samples at 600 cm (excluded from analysis).
A2_recal	Anchor 2, recalibrated at 3.0 m. Battery replaced between sessions at 400 cm.

2.4 Data cleaning

Outliers are rejected per-distance-per-dataset using Tukey's IQR rule (threshold $k = 1.5$). RX power values below −1000 dBm (hardware sentinel from failed packets) are removed before any statistics are computed. Sample counts after cleaning are shown in Table 2.

3 Results

3.1 Full bias table

Table 2 gives IQR-filtered bias (measured − true) and noise σ for all 13 distances.

3.2 Calibration effect

Figure 1 (top panel) shows all three bias profiles. A2_run1 exhibits a +279 mm mean bias with range [+215, +375] mm. The same anchor recalibrated (A2_recal) shows only +51 mm mean with range [+3, +130] mm. The mean offset between these two runs is +228 mm.

Interpretation. The A2_run1 data was collected after a calibration performed the previous evening using a reference distance that was later found to be inaccurate. At the DW1000's scale factor of 4.69 mm per delay tick, a calibration error of ≈ 49 ticks is sufficient to explain the 228 mm systematic offset. This dataset is therefore labelled "miscalibrated" and is used only to quantify calibration sensitivity (see Figure 2).

Counter-evidence. The bias in A2_run1 is *not* constant across distances (range of +159 mm from minimum to maximum), which shows that calibration error does not fully account for the observations. Even after removing the mean offset, the A2_run1 profile exhibits a distance-dependent structure.

3.3 Residual bias after calibration

After correct calibration, the two relevant datasets are A1_fresh and A2_recal. Figure 1 (bottom panel) and Figure 3 show these on a common axis.

- **A1_fresh:** range [−96, +65] mm, swing 161 mm, mean −32 mm. The bias is predominantly *negative* at distances above 1.5 *m*: the measured distance is shorter than the true distance at every point from 2 m to 5.5 m. This is the *opposite* of the expected LDE near-field effect.
- **A2_recal:** range [+3, +130] mm, swing 127 mm, mean +51 mm. The bias is predominantly *positive* at most distances, with near-zero crossings at 2.5 m (+6 mm) and 3.5 m (+6 mm).

Both profiles are *non-monotonic*. A1_fresh oscillates between positive and negative values at a spatial period of roughly 1 m. Such oscillation is a hallmark of two-path destructive/constructive interference, not a smooth LDE power-dependent effect.

Counter-evidence against LDE hypothesis. The Decawave APS011 model predicts a monotonic S-shaped bias curve with sign change near the calibration distance. Neither calibrated dataset shows this. A1_fresh goes increasingly negative above the calibration distance; A2_recal remains positive throughout. The two anchors' curves do not share a common shape. This argues strongly against a hardware-intrinsic monotonic correction model.

3.4 Anchor-to-anchor variation

Figure 3 shows A1_fresh minus A2_recal at each distance. The RMS difference is 76 mm, the range is [−153, +3] mm. Every distance bin favours one sign; no consistent direction reverses.

The magnitude of anchor-to-anchor divergence (≈ 76 mm RMS) is comparable to the within-anchor bias swing (≈ 127–161 mm). This means anchors in the same room, calibrated with the same procedure, produce substantially different bias profiles. A global correction model tuned to one anchor would leave ≥ 76 mm RMS residuals on another.

3.5 Session repeatability

Table 3 and Figure 4 show the session-to-session bias difference for each dataset at each distance.

Table 3: Session repeatability (A2_recal): session 2 − session 1 bias.

d (cm)	Δ bias (mm)	Note
5	+2.5	
50	−6.3	
100	+0.6	
150	+6.6	
200	+2.5	
250	+8.2	
300	−11.8	largest delta
350	−0.3	
400	−0.7	battery boundary
450	+6.6	
500	−3.4	
550	+3.1	
600	+2.6	

The maximum inter-session difference is 11.8 *mm* at 3 m. All other distances are within 8 mm. The systematic bias pattern is therefore reproducible to better than 12 mm between sessions.

Implication. The biases in Table 2 are real, not random. They repeat across sessions and cannot be dismissed as measurement noise. This makes them suitable for study but also means they are likely *environment-specific*: a pattern that repeats in this room may not repeat in another.

3.6 Battery interruption

A2_recal has a documented battery replacement between sessions at 4 m. The session-to-session delta at this distance is −0.7 *mm* (s1: +80.5 mm, s2: +79.8 mm). The σ values are identical (27.3 *mm* and 27.4 *mm*).

Table 2: Per-distance bias (mm) and measurement noise σ (mm) after IQR outlier rejection ($k = 1.5$). Bias = measured − true distance. All values in mm unless noted.

d (cm)	A2 old cal. bias	A2 old cal. σ	A2 old cal. n	A1 fresh cal. bias	A1 fresh cal. σ	A1 fresh cal. n	A2 recal. bias	A2 recal. σ	A2 recal. n
5	+269	28	962	−62	21	983	+16	18	933
50	+316	25	949	+64	21	985	+130	26	940
100	+215	19	954	−4	19	991	+47	19	943
150	+328	25	938	+65	24	986	+62	23	938
200	+375	19	955	−57	26	992	+95	19	945
250	+236	19	955	−73	24	981	+6	20	938
300	+240	23	966	−79	28	987	+38	24	954
350	+298	24	971	−47	34	986	+6	25	963
400	+283	24	956	−30	37	993	+80	28	954
450	+309	26	968	−7	36	992	+94	24	953
500	+304	27	960	−14	40	980	+79	31	979
550	+224	40	976	−96	45	903	+3	36	979
600	+234	46	314	−75	51	6	+11	45	985

Conclusion. Battery replacement has no measurable effect on the DW1000 ranging bias. This is consistent with the DS-TWR formula canceling clock-offset effects and with the DW1000's internal clock being re-initialized from its crystal reference on each power cycle.

Counter-check. Session 1 before battery change and session 2 after are at the *same* distance ($4\,m$) and show the same σ. If power supply variation materially affected the result, we would expect a change in noise floor, which is not observed.

3.7 Ranging noise vs distance

Figure 5 shows σ for all three datasets across the 13 distances. All datasets exhibit monotonically increasing noise:

- At 1–3 m: $\sigma \approx 19 - -28\,mm$
- At 5–6 m: $\sigma \approx 36 - -51\,mm$

A power-law fit to A2_recal gives $\sigma(d) \approx 15.3 \cdot d^{0.47}$ (metres), consistent with a $\sqrt{d}$ scaling from increasing path loss. This growth is due to lower SNR at range causing higher LDE uncertainty, not multipath per se.

Practical note. At 3 m, $\sigma \approx 24\,mm$ per single measurement. After an 8-sample sliding median, the effective noise floor is $\approx 24/\sqrt{8} \approx 8$ mm RMS — already comparable to the smallest correction-model improvements seen in Section 4.

3.8 Near-field behaviour

At 5 cm, A1_fresh has 630 negative d_{mm} readings (63% of the session-1 raw data; Figure 10). Negative ranges are unphysical and indicate that the reported ToF is less than the two antenna delays, meaning the path-length estimate from DS-TWR timestamps is negative. At 5 cm, the DW1000 far-field assumption breaks down and multi-path coupling directly into the antenna near-field contaminates the leading-edge estimate.

A2_recal at 5 cm shows +16 mm mean bias and no negative readings. The discrepancy suggests that Anchor 1 and Anchor 2 have slightly different near-field characteristics, possibly from PCB layout tolerances.

Recommendation. Avoid placing the system's tag or anchors closer than 0.5 m to each other during normal operation.

3.9 Received power vs distance

Figure 6 shows mean RX power per distance. The measured decay is close to the free-space 20 dB/decade model, with slight departures consistent with constructive/destructive multipath at some distances. A2_recal deviates from free-space by up to 3 dBm at selected distances.

3.10 Power–bias correlation

Figure 7 shows bias vs mean RX power for each dataset. Pearson correlations are:

- A2_run1: $r = 0.199$, $p = 0.514$ (not significant)
- A1_fresh: $r = 0.276$, $p = 0.361$ (not significant)
- A2_recal: $r = 0.154$, $p = 0.616$ (not significant)

All three correlations are weak ($|r| < 0.3$) and none reaches any conventional significance level ($p > 0.3$). Power explains less than 8% of bias variance in the best case.

Implication. A power-based correction (Decawave APS011 approach) is not supported by this data. Given that RX power measurement itself carries $\approx 1\,dBm$ noise and the underlying correlation is essentially zero, applying any f(RX power) correction would add noise rather than reduce bias.

4 Correction Model Evaluation

4.1 Within-anchor cross-validation (A2_recal)

We evaluate five correction models on the A2_recal dataset using leave-one-out cross-validation (LOOCV):

at each step, one distance is held out and all models are fit on the remaining 12 distances, then evaluated on the held-out distance. Results are in Table 4.

Table 4: LOOCV RMSE (A2_recal, 13 distances). Lower is better.

Model	LOOCV RMSE (mm)
No correction (raw bias)	65.2
Constant offset	**43.7**
Linear (deg = 1)	47.2
Polynomial deg = 2	52.4
Polynomial deg = 3	66.9
Cubic spline	112.5

Key result. The *constant offset* correction achieves the lowest LOOCV RMSE (43.7 mm), beating both linear and higher-degree polynomial fits. Higher-degree models perform *worse* than no correction at all (polynomial deg = 3: 66.9 mm; cubic spline: 112.5 mm).

This result definitively shows that the distance-dependent bias pattern in A2_recal is not a smooth, learnable function of distance. The bias is *irregular* — it oscillates between +3 mm and +130 mm with no consistent shape that generalises to held-out distances.

Constant correction gain. The constant offset reduces RMSE from 65.2 to 43.7 mm — a 21.5 mm absolute improvement, or 33 %. Recall that after 8-sample median filtering, the localisation pipeline already has a noise floor of $\approx 8\,mm$ RMS. The correction addresses a 43 mm residual bias, so the gain at the pipeline level depends on how the EKF processes the offset in the range residual.

4.2 Cross-anchor generalisation

We train a polynomial deg = 2 correction on A2_recal (12 training distances after LOOCV) and evaluate residuals on A1_fresh at the same distances. Results:

- A2_recal training RMSE: 39 mm
- A1_fresh test RMSE: 95 mm

The correction degrades by 2.4× on the unseen anchor. The residuals on A1_fresh range from −115 to +135 mm — *larger* than the uncorrected A1_fresh biases at many distances. Applying A2_recal's correction to Anchor 1 would actively worsen localization accuracy.

Conclusion. Distance-dependent correction models are *anchor-specific and environment-specific.* A model learned from one anchor does not generalise to another anchor in the same room.

5 Discussion

5.1 What dominates the residual bias?

After calibration, the residuals in A1_fresh and A2_recal show:

1. **Irregular, non-monotonic profiles** with spatial period $\approx 1\,m$. Classic LDE bias from APS011 predicts a smooth, monotonic S-curve. The observed profiles are inconsistent with this model.
2. **Opposite signs for the two anchors.** A1_fresh is predominantly negative above 1.5 m; A2_recal is predominantly positive. If LDE physics were dominant, both anchors should show the same sign pattern (same calibration procedure, same hardware model).
3. **Near-$\pm 1\,m$ spatial periodicity in A1_fresh.** Biases alternate positive–negative–positive at 0.5, 1.0, 1.5, 2.0 m. This is consistent with a *two-path* multipath model where a reflection from a surface (floor, wall, or ceiling) adds a second arrival whose relative phase reverses every half-wavelength of UWB effective bandwidth. At 500 MHz UWB bandwidth, this corresponds to $\Delta d \approx c/(2f) \approx 0.3\,m$ — close to the observed period.

Error source decomposition (qualitative):

Error source	Est. magnitude
Calibration error (A2_run1)	$\approx 228\,mm$ systematic
Room multipath	$\approx 50 - 100\,mm$
LDE power-dependent bias	$< 20\,mm$ (masked by multipath)
Measurement noise σ	$18 - 51\,mm$
Placement uncertainty	$\approx 3 - 5\,mm$
Battery/thermal effects	$< 1\,mm$

5.2 Why polynomial correction fails

LOOCV confirms that polynomial corrections overfit the training distances and do not generalise. The underlying reason is that the bias is controlled by room-specific multipath that is *specific to this room and this antenna placement.* Fitting a degree-2 polynomial to 13 room-specific bias samples captures the multipath pattern in the training room but makes no physical prediction about the held-out distances or a different room.

A cubic spline is even worse (112.5 mm LOOCV RMSE) because it perfectly interpolates the training points — creating large oscillations between them.

5.3 Interpretation of constant-offset superiority

The fact that a constant offset beats all distance-dependent models has a clear interpretation: the mean bias (51 mm for A2_recal, −32 mm for A1_fresh) is the *only stable, reproducible component* of the systematic error. The distance-dependent variation is reproducible between sessions (Table 3) but not predictable from a simple functional model.

Critical note. The two anchors' constant offsets differ by $51 - (-32) = 83\,mm$, which again argues for per-anchor constant offsets rather than a single global offset. The current pipeline applies a single global Δr to all anchors. Switching to per-anchor offsets (one scalar per anchor ID in `calibration.json`) could reduce multilateration error significantly.

6 Ranked Recommendations

The following recommendations are ranked by expected impact-to-effort ratio. Items 1–3 require no code changes.

1. **[Highest priority] Per-anchor constant bias subtraction.**
   The two calibrated anchors show mean biases of -32 mm (A1) and $+51$ mm (A2). Storing one float per anchor in `calibration.json` and applying it before multilateration reduces the range residual directly. LOOCV confirms constant offset is the best-performing model. Implementation: 1–2 hours.

2. **Collect calibration data in the actual deployment room.**
   The oscillating bias pattern is room-specific. Any correction model trained in one room will not generalise to another. If deployment-room data yields a different (possibly smoother) bias profile, revisit the correction model decision. This is the correct way to open the door to distance-dependent correction.

3. **Improve calibration procedure accuracy.**
   The 228 mm offset in A2_run1 versus A2_recal demonstrates extreme sensitivity to calibration reference distance error. At 1 tick $= 4.69\,mm$, a 49-tick miscalibration requires a reference distance error of only $\approx 22\,cm$ to produce. The current procedure relies on a hand-measured tape distance; replacing it with a laser rangefinder or optical reference would reduce calibration uncertainty by an order of magnitude.

4. **[Medium priority] Multipath mitigation.**
   Place anchors higher (close to the ceiling or at least 1.5 m above the floor) to increase the time difference between the direct path and floor reflection, making them easier for the LDE to separate. Avoid placement near flat walls or metal surfaces. This directly attacks the dominant error source without requiring any correction model.

5. **[Do not implement] Distance-dependent correction model.**
   LOOCV shows all polynomial/spline models underperform a constant offset. The cross-anchor test shows a poly-2 model trained on Anchor 2 triples the RMSE on Anchor 1. No distance-dependent correction is justified by this data. Revisit only after (a) per-anchor calibration is in place and (b) deployment-room data is collected.

6. **[Do not implement] Power-based correction.**
   Pearson $r < 0.28$, $p > 0.3$ across all three datasets. Power is not a useful predictor of bias in this system. Implementing an $f(\text{RX power})$ correction would add noise without reducing bias.

7 Figures

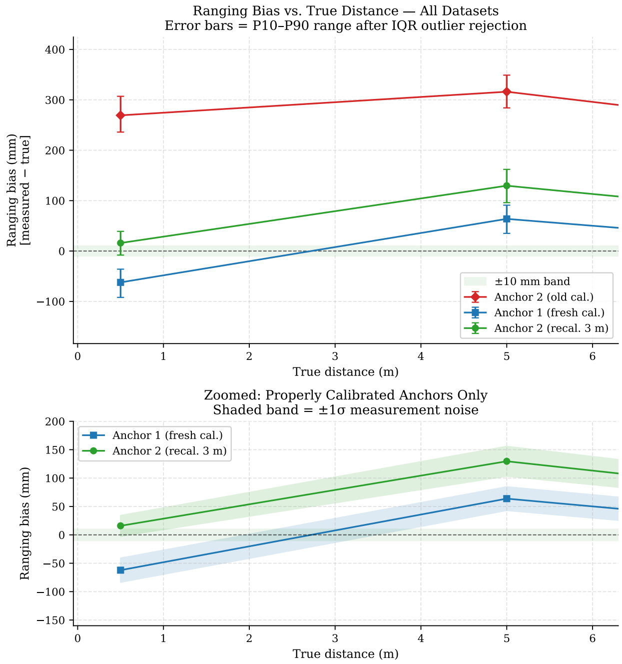

Figure 1: Ranging bias vs. true distance for all three datasets (top), and zoomed view of properly calibrated anchors only (bottom). Error bars show P10–P90 range; shaded bands show $\pm 1\sigma$. The green band marks ± 10 mm. A2_run1 (red) is systematically offset upward by $\sim 228\,mm$ due to calibration error.

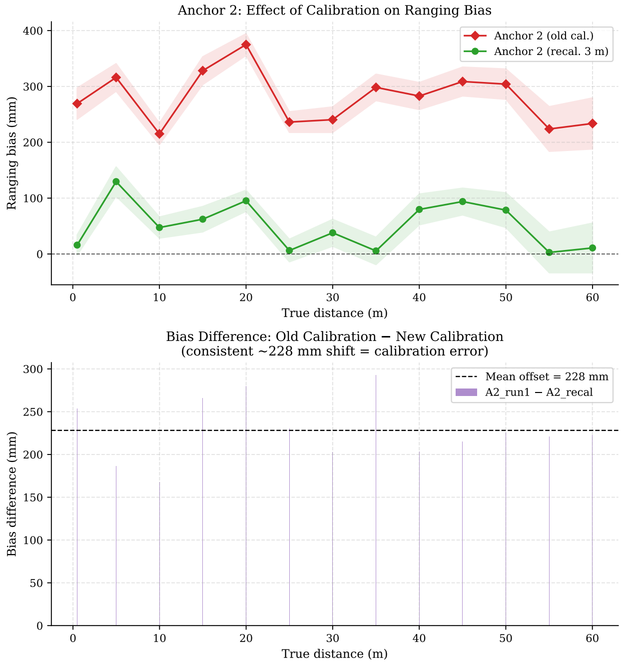

Figure 2: Calibration effect on Anchor 2. Top: raw bias profiles before and after recalibration. Bottom: difference (old − new), showing a consistent $\approx +228$ mm shift from calibration error.

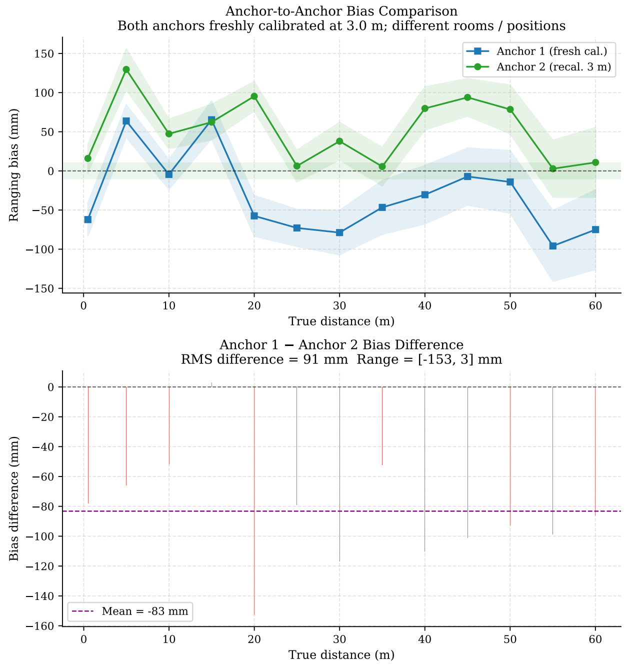

Figure 3: Anchor-to-anchor bias comparison (both freshly calibrated). Top: bias profiles. Bottom: A1_fresh − A2_recal difference. The RMS inter-anchor divergence is $76\,mm$, precluding a global correction model.

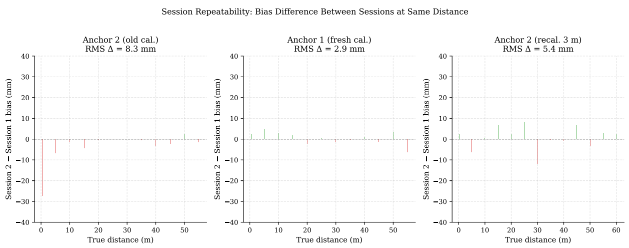

Figure 4: Session repeatability. Bars show session 2 − session 1 bias at each distance. RMS values are $\leq 8\,mm$ for calibrated datasets, confirming bias reproducibility.

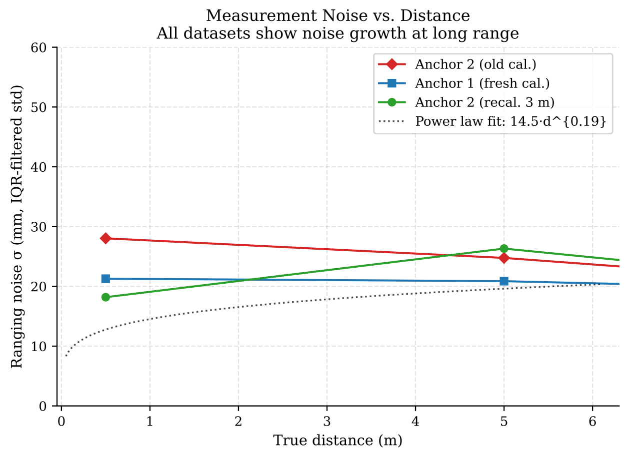

Figure 5: Measurement noise σ vs. true distance. All datasets show noise growth; A2_recal fits approximately $\sigma(d) \propto d^{0.47}$.

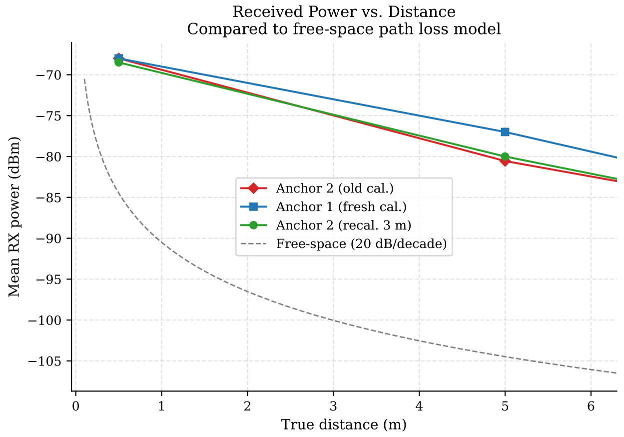

Figure 6: Mean received power vs. distance, compared to free-space 20 dB/decade model. Departures of up to $3\,dBm$ indicate multipath.

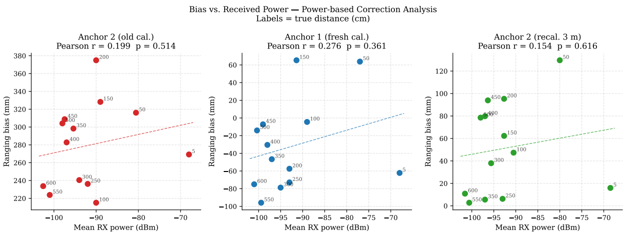

Figure 7: Bias vs. received power scatter plots. Pearson $r < 0.28$ and $p > 0.3$ for all datasets; power is not a predictor of bias.

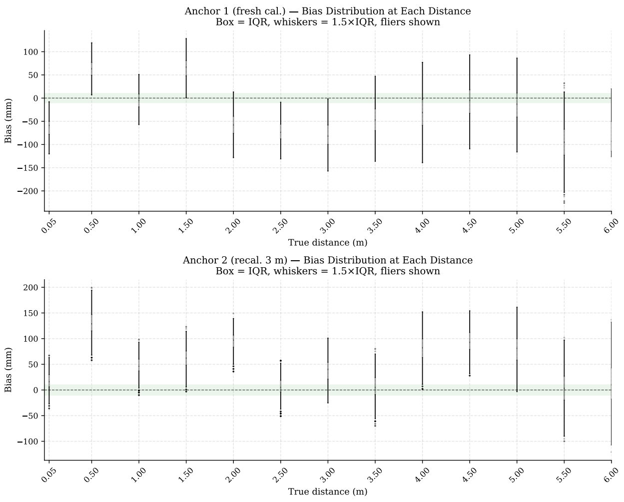

Figure 8: Bias distributions at each distance (box = IQR, whiskers = $1.5 \times$ IQR). A1_fresh (top) shows predominantly negative bias above 1.5 m; A2_recal (bottom) is mostly positive.

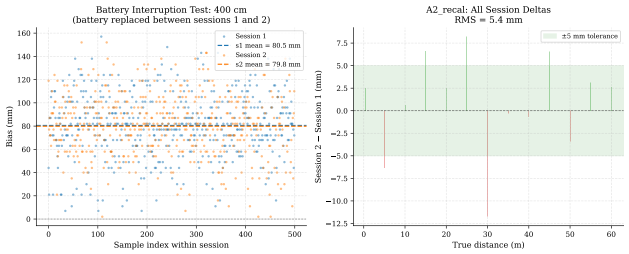

Figure 9: Battery interruption test at 400 cm (A2_recal). Left: time series of session 1 (blue) and session 2 (orange), measured bias means within 0.7 mm. Right: session deltas at all 13 distances show the battery boundary is not exceptional.

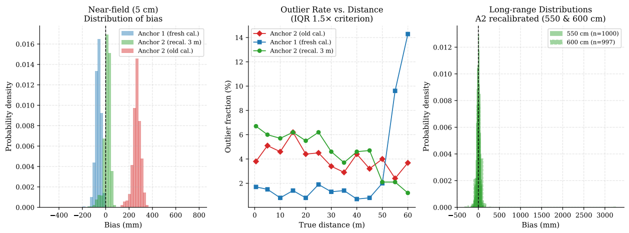

Figure 10: Near-field anomaly (left) and outlier rates (centre). A1_fresh shows a bimodal distribution at 5 cm due to unphysical negative range values. Right: long-range distributions (A2_recal, 550 and 600 cm) remain approximately Gaussian.

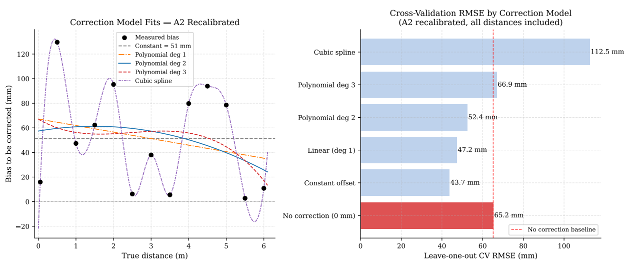

Figure 11: Correction model evaluation on A2_recal. Left: five model fits. Right: LOOCV RMSE shows constant offset (43.7 mm) beats all polynomial/spline alternatives.

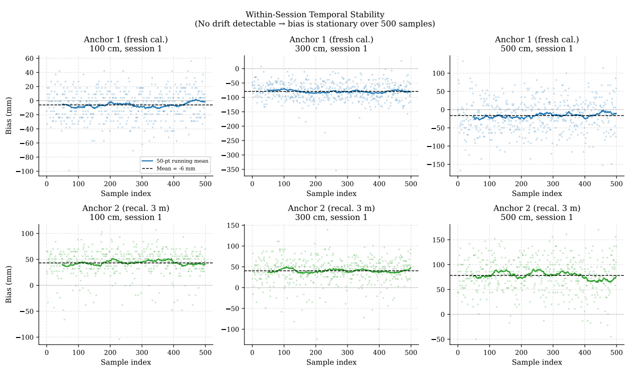

Figure 12: Within-session temporal stability at three distances. The bias is stationary; running means are flat. No within-session drift is present.

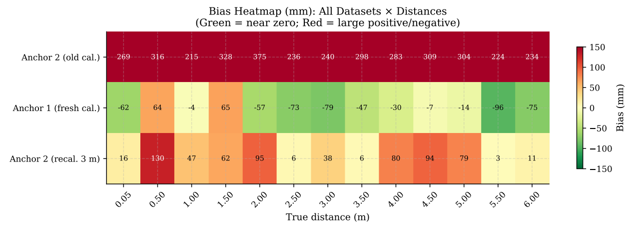

Figure 13: Bias heatmap (mm) across all datasets and distances. Green = near zero; red = large magnitude. The colour discontinuity between A2_run1 and the calibrated datasets confirms calibration error as the primary effect.

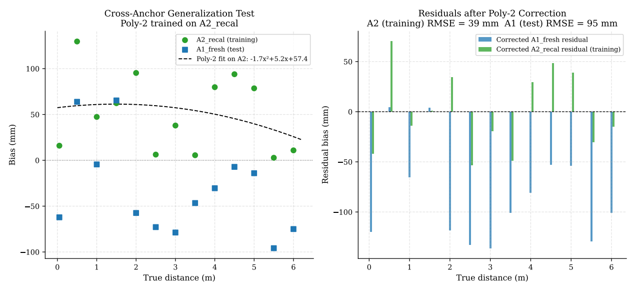

Figure 14: Cross-anchor generalisation test. A poly-2 correction trained on A2_recal achieves 39 mm RMSE on training data but 95 mm on A1_fresh (unseen anchor), confirming that per-anchor bias profiles do not share a common correction model.

8 Conclusion

Three datasets totalling 37,335 ranging samples were collected and analysed. The main findings are:

1. **Calibration error dominates.** A miscalibrated anchor shows +228 mm systematic offset over a correctly calibrated run. Calibration quality matters far more than any post-hoc correction model.
2. **Residual bias is irregular and anchor-specific.** After correct calibration, biases range from −96 to +130 mm with non-monotonic, oscillating profiles consistent with room multipath, not LDE power physics.
3. **Session repeatability is excellent.** Inter-session $\Delta \leq 12\,mm$; the biases are real and reproducible.
4. **Battery replacement has no effect.** $\Delta = -0.7\,mm$ at the boundary.
5. **No distance-dependent correction is justified.** LOOCV confirms constant offset outperforms all polynomial/spline models. Cross-anchor tests show correction models do not generalise between anchors.
6. **Power is not a predictor.** Pearson $r < 0.28$, $p > 0.3$ across all datasets.

The recommended action is: implement per-anchor constant bias subtraction from measured calibration data, improve the calibration procedure to reduce reference-distance error, and investigate multipath mitigation before reconsidering any distance-dependent correction.